ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

MACHINE VISION

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 3rd year Machine Vision paper 1 · practice a.

Written by Academic Central for practice, following undergraduate machine vision and image processing courses for mechatronics students (topics as in the image processing and robotic vision content of South African engineering programmes, e.g. University of Pretoria EAI 733, and standard introductory machine vision courses). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any well-reasoned design in QUESTION 5.1 and answers that differ only through rounding.

QUESTION 1

1.1.1 h_image = f × H / Z = 8 mm × 0.5 / 2 ✓ = 2 mm ✓ (2)

1.1.2 2 mm / 4 µm ✓ = 500 pixels ✓ (2)

1.2 u = f_x X/Z + c_x = 800 × 0.1 + 320 ✓ = 400 ✓ (4)
v = f_y Y/Z + c_y = 800 × (−0.05) + 240 ✓ = 200 ✓

1.3 HFOV = 2 $\tan^{-1}$(w / 2f) ✓ = 2 $\tan^{-1}$(3.2 / 8) ✓ = 43.60° ✓ (3)

1.4 400 mm / 1 mm = 400 defect widths ✓ (3)
× 2 pixels each ✓ = 800 pixels (choose a standard sensor of at least 800 pixels wide, e.g. 1 024 or 1 280) ✓

1.5 Radial distortion: straight lines appear curved (barrel or pincushion), more strongly towards the edge of the image ✓ (3)
Calibration finds the focal length, principal point and distortion coefficients ✓, so the image can be corrected and pixels converted accurately into real-world measurements ✓

1.6 A rolling shutter exposes the rows one after another, so a moving part is skewed or bent in the image ✓ (3)
A global shutter exposes all the pixels at the same instant ✓, so moving parts are captured without distortion and can be measured correctly ✓

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QUESTION 2

2.1 640 × 480 × 3 bytes ✓ = 921 600 bytes (900 KB) ✓ (2)

2.2 0.299 × 200 + 0.587 × 100 + 0.114 × 50 = 59.8 + 58.7 + 5.7 ✓ = 124.2, so 124 ✓ (2)

2.3 g = (f − 50) × 255 / (150 − 50) ✓✓ (3)
g(100) = 50 × 2.55 = 127.5, so 128 ✓

2.4 It tries every possible threshold on the histogram ✓, splitting the pixels into two classes each time ✓, and chooses the threshold that gives the smallest within-class variance (the largest between-class variance) ✓ (3)

2.5 Correct binary image ✓✓✓ (one mark per four correct pixels, 127 → 0 and 128 → 1) (3)

0	1	1	0
1	1	1	0
0	1	0	0
0	0	1	1

2.6 HSV keeps the colour (hue) separate from the brightness (value) ✓, so the hue of an object stays almost the same when the lighting gets brighter or darker, and one hue range finds it ✓ (2)

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QUESTION 3

3.1 Neighbourhood rows 1 to 3, columns 1 to 3: each row is 10, 10, 80 ✓ (3)
Sum = 3 × 100 = 300 ✓
300 / 9 = 33.33 ✓

3.2.1 Neighbourhood rows 1 to 3, columns 0 to 2: eight 10s and one 250 ✓ (3)
Sorted: 10, 10, 10, 10, 10, 10, 10, 10, 250 ✓
Median = 10: the noise is removed completely ✓

3.2.2 (8 × 10 + 250) / 9 = 36.67 ✓ (2)
The mean smears the spike into the neighbours; the median filter is better for this "salt-and-pepper" noise ✓

3.3.1 Neighbourhood rows 0 to 2, columns 1 to 3: each row is 10, 10, 80 ✓ (4)
$G_x = 1(80 - 10) + 2(80 - 10) + 1(80 - 10)$ ✓✓ = 280 ✓

3.3.2 Row 2 − row 0 = 0 in every column, so $G_y = 0$ ✓✓ (2)

3.3.3 Magnitude = $\sqrt{(280^2 + 0^2)} = 280$; direction = $\tan^{-1}(0/280) = 0°$ ✓ (2)
The intensity changes strongly from left to right (horizontal gradient), which shows a vertical edge ✓

3.4 Edge detectors use derivatives, which strongly amplify noise ✓; smoothing first removes noise so that false edges are not detected ✓ (2)

3.5 Smooth with a Gaussian ✓ (4)
Find the gradient magnitude and direction (e.g. Sobel) ✓
Non-maximum suppression to thin the edges to one pixel ✓
Double threshold and hysteresis to keep strong edges and the weak edges connected to them ✓

3.6 Each edge pixel votes for all the lines that could pass through it, using the line form $\rho = x \cos\theta + y \sin\theta$ ✓ (3)
The votes are added up in an accumulator array of $(\rho; \theta)$ ✓
Peaks in the accumulator are the lines supported by many edge pixels ✓

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QUESTION 4

4.1 Erosion: a pixel stays 1 only if the whole structuring element fits inside the object; objects shrink ✓ (6)
Dilation: a pixel becomes 1 if the element touches the object; objects grow ✓
Opening = erosion then dilation ✓; removes small noise and thin connections while keeping the size of larger objects ✓
Closing = dilation then erosion ✓; fills small holes and gaps in objects ✓

4.2.1 Only the centre pixel (2, 2) remains 1; everything else, including the noise pixel at (0, 0), becomes 0 ✓✓✓ (3)

4.2.2 The 3 × 3 square at rows 1 to 3, columns 1 to 3 is restored, and (0, 0) stays 0 ✓
The opening removed the noise pixel without changing the square ✓ (2)

4.3 Labelling scans the binary image and gives every separate object (group of connected 1-pixels) its own number ✓
4-connectivity: pixels are neighbours only if they share an edge (up, down, left, right) ✓
8-connectivity: diagonal pixels are also neighbours, so objects touching at corners join ✓ (3)

4.4 Area = 5 pixels ✓
Centroid row = (1 + 1 + 2 + 2 + 3) / 5 = 1.8 ✓
Centroid column = (1 + 2 + 1 + 2 + 2) / 5 = 1.6 ✓
Centroid = (1.8; 1.6) ✓ (4)

4.5 $4\pi \times 1\,200 / 150^2$ ✓ = 0.67; a perfect circle gives 1 ✓ (2)

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QUESTION 5

5.1 Lighting: e.g. a backlight behind the bottles or a diffuse light from above, enclosed to block daylight ✓
Camera: a global-shutter industrial camera placed so the neck of every bottle is in view ✓
Trigger: a photo-eye sensor triggers the camera (and a strobe light) when a bottle is in position ✓
Processing: region of interest at the neck, threshold, count the cap pixels or measure the blob height, compare with limits ✓✓
Output: a pass or fail signal to the PLC, which tracks the bottle (encoder or timer) and fires the reject air jet when the bottle reaches it ✓ (6)

5.2 A small template image is slid over the image, and at each position the normalised correlation is calculated; the highest score (close to 1) is the best match ✓
Any TWO limitations: it fails when the object is rotated or scaled; slow for large images; affected by partial hiding (occlusion); affected by large changes in appearance ✓✓ (3)

5.3.1 $Z = fB / d$ ✓ = 700 × 0.12 / 35 ✓ = 2.4 m ✓ (3)

5.3.2 d = 34: Z = 2.47 m; d = 36: Z = 2.33 m ✓ (2)
A 1-pixel error gives about a 7 cm error at 2.4 m; the depth error grows with distance, so far objects are measured less accurately ✓

5.4 Δcolumn = 400 − 320 = 80 pixels, Δrow = 300 − 240 = 60 pixels ✓ (3)
x = 80 × 0.5 = 40 mm ✓ y = 60 × 0.5 = 30 mm ✓

5.5 Advantage (any one): copes with natural variation in defects and backgrounds; can learn complex defects that are hard to describe with rules ✓ (3)
Disadvantage (any one): needs a large set of labelled images; needs powerful hardware; harder to explain why it made a decision ✓✓

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TOTAL: 100

END OF MARKING GUIDELINE